

Example: p -adic fields

1 p -adic fields

Construction 1. Let K be a number field and $\mathfrak{p}$ be a prime ideal of the ring of integers $\mathcal{O}_K$ of K . Considering the localization $(\mathcal{O}_K)_{\mathfrak{p}}$ of $\mathcal{O}_K$ at $\mathfrak{p}$, which is a discrete valuation ring, denote by $v_{\mathfrak{p}} : K^{\times} \rightarrow \mathbb{Z}$ the corresponding discrete valuation. The p -adic absolute value on K associated to $\mathfrak{p}$ is defined as

$$|x|_{\mathfrak{p}} := N(\mathfrak{p})^{-v_{\mathfrak{p}}(x)}, \quad \forall x \in K,$$

where $N(\mathfrak{p}) := \#(\mathcal{O}_K/\mathfrak{p})$ is the norm of $\mathfrak{p}$.

The completion of K with respect to the p -adic absolute value $|\cdot|_{\mathfrak{p}}$ is denoted by $K_{\mathfrak{p}}$, called the p -adic field.

We just focus on the case $K = \mathbb{Q}$ and $\mathfrak{p} = (p)$ for a prime number p in the following.

Example 2. Let p be a prime number. For every $r \in \mathbb{Q}$, we can write r as $r = p^n \frac{a}{b}$, where $n \in \mathbb{Z}$ and $a, b \in \mathbb{Z}$ are integers not divisible by p . The p -adic absolute value on $\mathbb{Q}$ is defined as

$$|r|_p := p^{-n}.$$

The p -adic field $\mathbb{Q}_p$ can be described concretely as follows:

$$\mathbb{Q}_p = \left\{ \sum_{i=n}^{+\infty} a_i p^i \mid n \in \mathbb{Z}, a_i \in \{0, 1, \dots, p-1\} \right\}.$$

For $x = \sum_{i=n}^{+\infty} a_i p^i \in \mathbb{Q}_p$ with $a_n \neq 0$, its p -adic absolute value is given by $|x|_p = p^{-n}$. The operations of addition and multiplication on $\mathbb{Q}_p$ are defined similarly as those on decimal expansions.

Easily see that $|\mathbb{Q}_p^{\times}|_p = p^{\mathbb{Z}}$ and $\mathcal{K}_{\mathbb{Q}_p} \cong \mathbb{F}_p$.

Unlike the field of real numbers $\mathbb{R}$, the p -adic field $\mathbb{Q}_p$ has many finite extensions.

Proposition 3. The field $\mathbb{Q}_p$ has algebraic extensions of arbitrarily large degree.

Proof. Since there are infinitely many irreducible monic polynomials over the finite field $\mathbb{F}_p$, consider any lift of such an irreducible monic polynomial to a monic polynomial with coefficients in $\mathbb{Z}_p$. If the lift is not irreducible over $\mathbb{Q}_p$, then the factorization of the lift gives a nontrivial factorization of its reduction modulo p since the factors can be chosen to be monic and have coefficients in $\mathbb{Z}_p$, which contradicts the irreducibility of the original polynomial over $\mathbb{F}_p$. Thus, the lift is irreducible over $\mathbb{Q}_p$.

On the other hand, note that $|\mathbb{Q}_p^{\times}|_p = p^{\mathbb{Z}}$. It follows that $f(T) = T^n - p$ is irreducible over $\mathbb{Q}_p$ for every integer $n \geq 1$. Otherwise, suppose we have a monic factorization $f(T) = g(T)h(T)$ with $g(T), h(T) \in \mathbb{Z}_p[T]$ and $\deg g, \deg h < n$. Then by considering the reduction modulo p , we have $g(0), h(0) \equiv 0 \pmod{p}$. It follows that $|f(0)|_p = |g(0)h(0)|_p \leq p^{-2}$, which contradicts $|f(0)|_p = |p|_p = p^{-1}$. $\square$

Let

$$R := \bigcup_{n=1}^{+\infty} \mathbb{Q}_p(\mu_{p^n})$$

and

$$K := \bigcup_{n=1}^{+\infty} \mathbb{Q}_p\left(\bigcup_{p \nmid m} \mu_m\right).$$

Yang: R is a totally ramified extension and K is the maximal unramified extension.

2 Completion

Proposition 4. The algebraic closure $\overline{\mathbb{Q}_p}$ of $\mathbb{Q}_p$ is not complete with respect to the extension of the p -adic absolute value $|\cdot|_p$.

Proof. Let $(a_n)_{n \geq 1}$ be a sequence in $\overline{\mathbb{Q}_p}$ such that

$$[\mathbb{Q}_p(\{a_n\}) : \mathbb{Q}_p] = +\infty.$$

Set $s_1 = 0$ and

$$s_n = s_{n-1} + p^{k_n} a_n$$

for a suitable $k_n \in \mathbb{N}$ such that

$$|p^{k_n} a_n|_p \leq \frac{|p^{k_{n-1}} a_{n-1}|_p}{2}, \quad |p^{k_n} a_n|_p < |s_{n-1} - \sigma(s_{n-1})|_p \quad \forall \sigma \in \text{Gal}(\overline{\mathbb{Q}_p}/\mathbb{Q}_p) \text{ with } \sigma(s_{n-1}) \neq s_{n-1}.$$

Then the sequence $(s_n)_{n \geq 1}$ is a Cauchy sequence. Let $s \in \widehat{\overline{\mathbb{Q}_p}}$ be the limit of $(s_n)_{n \geq 1}$. We have $|s - s_{n-1}|_p = |p^{k_n} a_n|_p < |s_{n-1} - \sigma(s_{n-1})|_p$ for all $n \geq 1$ and all $\sigma \in \text{Gal}(\overline{\mathbb{Q}_p}/\mathbb{Q}_p)$ with $\sigma(s_{n-1}) \neq s_{n-1}$. By Krasner's lemma ([Theorem 12](#)), it follows that $s_n \in \mathbb{Q}_p(s)$ for all $n \geq 1$. Thus, we have

$$[\mathbb{Q}_p(s) : \mathbb{Q}_p] \geq [\mathbb{Q}_p(\{a_n\}) : \mathbb{Q}_p] = +\infty,$$

which implies that $s \notin \overline{\mathbb{Q}_p}$. □

Construction 5. Let p be a prime number. The field $\mathbb{C}_p$ of p -adic complex numbers is defined as the completion of the algebraic closure of $\mathbb{Q}_p$ with respect to the unique extension of the p -adic absolute value $|\cdot|_p$ on $\mathbb{Q}_p$.

The field $\mathbb{C}_p$ is algebraically closed and complete with respect to $|\cdot|_p$ by [Proposition 9](#). By [Corollaries 10](#) and [11](#), we have

$$|\mathbb{C}_p^\times|_p = |\overline{\mathbb{Q}_p}^\times|_p = p^{\mathbb{Q}}, \quad \kappa_{\mathbb{C}_p} \cong \kappa_{\overline{\mathbb{Q}_p}} \cong \overline{\mathbb{F}_p}.$$

Proposition 6. The field $\mathbb{C}_p$ of p -adic complex numbers is not spherically complete.

Proof. Choose a strictly descending sequence $r_n \in |\mathbb{C}_p^\times|_p$ with $r_n \rightarrow r > 0$ as $n \rightarrow +\infty$. We inductively construct a collection of closed ball. Begin from r_0 , choose any nonempty closed ball E of radius r_0 . On E , choose two disjoint nonempty closed balls of radius r_1 and denote them by E_0 and E_1 . Inductively, for any binary sequence $a_1 a_2 \cdots a_n \in \{0, 1\}^n$, suppose that we have constructed the ball

$E_{a_1 \dots a_n}$ has radius r_n . On $E_{a_1 \dots a_n}$, choose two disjoint nonempty closed balls of radius r_{n+1} and denote them by $E_{a_1 \dots a_n 0}$ and $E_{a_1 \dots a_n 1}$. In this way, we obtain a collection of nonempty closed balls

$$\{E_{a_1 \dots a_n} \mid n \geq 0, a_1 \dots a_n \in \{0, 1\}^n\}.$$

For $a = a_1 a_2 \dots \in \{0, 1\}^{\mathbb{N}}$, set

$$E_a := \bigcap_{n=0}^{+\infty} E_{a_1 \dots a_n}.$$

Then each E_a is a closed ball of radius r . And for $a \neq b \in \{0, 1\}^{\mathbb{N}}$, we have $E_a \cap E_b = \emptyset$. Note that $\overline{\mathbb{Q}}$ is dense in $\mathbb{C}_p$. Since $\overline{\mathbb{Q}}$ is countable and $\{0, 1\}^{\mathbb{N}}$ is uncountable, there exists some $a \in \{0, 1\}^{\mathbb{N}}$ such that $E_a \cap \overline{\mathbb{Q}} = \emptyset$. In ultra-metric spaces, closed balls are also open. It follows that $E_a = \emptyset$. $\square$

Remark 7. For a concrete example, see [p-adic fields for beginners](#).

Construction 8. Let p be a prime number. *Yang: We construct the spherically complete p -adic field Ω_p . Yang: To be completed.*

Yang: does Ω_p has the same card as $\mathbb{R}$?

Appendix

Proposition 9. Let $\mathbf{k}$ be an algebraically closed non-archimedean field. Then its completion $\widehat{\mathbf{k}}$ is also algebraically closed.

Corollary 10. Let $\mathbf{k}$ be a non-archimedean field and $\widehat{\mathbf{k}}$ its completion. Then the residue field $\kappa_{\widehat{\mathbf{k}}} \cong \kappa_{\mathbf{k}}$ under the natural embedding $\mathbf{k}^\circ \hookrightarrow \widehat{\mathbf{k}}^\circ$.

Corollary 11. Let $\mathbf{k}$ be a non-archimedean field and $\widehat{\mathbf{k}}$ its completion. Then the valuation group $|\widehat{\mathbf{k}}^\times|$ of $\widehat{\mathbf{k}}$ is equal to the valuation group $|\mathbf{k}^\times|$ of $\mathbf{k}$.

Theorem 12 (Krasner's lemma). Let $\mathbf{k}$ be a complete non-archimedean field, and $\alpha, \beta \in \overline{\mathbf{k}}$. Denote by $\alpha_1, \alpha_2, \dots, \alpha_n$ the conjugates of α over $\mathbf{k}$ with $\alpha_1 = \alpha$. If

$$|\beta - \alpha| < |\alpha - \alpha_i|, \quad \forall i = 2, \dots, n,$$

then $\alpha \in \mathbf{k}(\beta)$.